

National Technical University of Athens

School of Electrical and
Computer Engineering (ECE)

10th semester

Control Systems of Electrical Machines

Simulation Assignment 4:
Improved Vector Control of an Induction Motor

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Introduction

This assignment studies improved vector control of an induction motor using indirect field orientation. It builds on the dynamic model and basic control structures implemented in the preceding simulation assignment, extending system operation to investigate the effects of parameter errors and the overall dynamic behaviour of the controller.

The supplied induction motor model is implemented in the rotating dq reference frame, with two-degree-of-freedom current and speed controllers. Rotor flux is estimated using a current model, which is used to calculate the field-orientation angle.

Figure (1) shows the overall simulation model, while Figure (2) shows the current-model subsystem for indirect field orientation developed in Simulation Assignment 3. The only implementation difference is that the rated current is taken as:

$$I_n = 5\ A \tag{1}$$

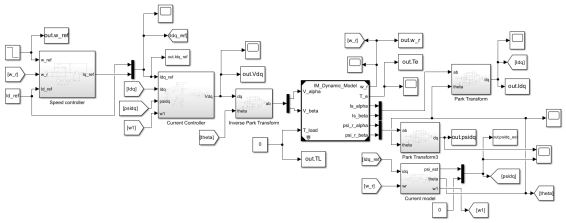


Figure 1: Complete IM model

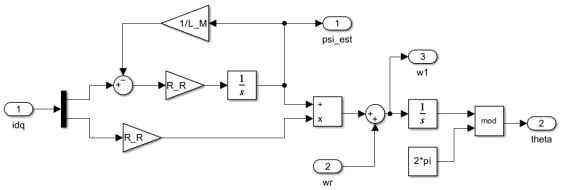


Figure 2: IFO current model subsystem

1 Sensitivity of the Current Model to Parameter Errors

This question examines the sensitivity of the current model to errors in rotor resistance R_R and magnetizing inductance L_M. The current model estimates rotor flux and therefore directly affects indirect field orientation. The rotor-flux estimate is based on:

$$\omega_1 = \omega_r + \frac{\widehat{R}_R i_q^{ref}}{\widehat{\psi}_R} \tag{2}$$

$$\theta_1 = \mathrm{mod}\left(\int \omega_1\, dt, 2\pi\right) \tag{3}$$

$$\widehat{\psi}_R = \int \widehat{R}_R \left(i_d^{ref} - \frac{\widehat{\psi}_R}{\widehat{L}_M}\right) dt \tag{4}$$

Both rotor resistance and magnetizing inductance directly enter the calculation of estimated flux and slip frequency. Deviations in these parameters therefore cause a field-orientation error that affects the dynamic behaviour of the overall control system, as shown by the following expressions:

$$\psi_d = \frac{L_M \psi_{ref}}{\widehat{L}_M} \frac{1 + \frac{R_R \widehat{L}_M}{\widehat{R}_R L_M}\left(\frac{\widehat{R}_R L_M i_q}{R_R \psi_{ref}}\right)^2}{1 + \left(\frac{\widehat{R}_R L_M i_q}{R_R \psi_{ref}}\right)^2} \qquad \text{and} \qquad \psi_q = L_M i_q \frac{1 - \frac{\widehat{R}_R L_M}{R_R \widehat{L}_M}}{1 + \left(\frac{\widehat{R}_R L_M i_q}{R_R \psi_{ref}}\right)^2} \tag{5}$$

Simulation Results

Three simulations were performed to investigate system sensitivity, using the following current-model parameter values:

1. $\widehat{R}_R = R_R$ and $\widehat{L}_M = L_M$
2. $\widehat{R}_R = 0.5R_R$ and $\widehat{L}_M = 1.5L_M$
3. $\widehat{R}_R = 2R_R$ and $\widehat{L}_M = 0.75L_M$

The first case uses the actual machine values, while the second and third cases use deliberately incorrect parameter estimates.

The reference for i_d is kept constant to produce rated rotor flux:

$$i_d^{ref} = I_{d,nom}\frac{\psi_N}{L_M} = \frac{0.9072}{0.3354} = 2.705 \text{ A} = 0.5313 \text{ p.u.} \tag{6}$$

Figures (3)–(6) show the speed response, the current component i_q and the rotor-flux components in the dq frame for the different parameter estimates.

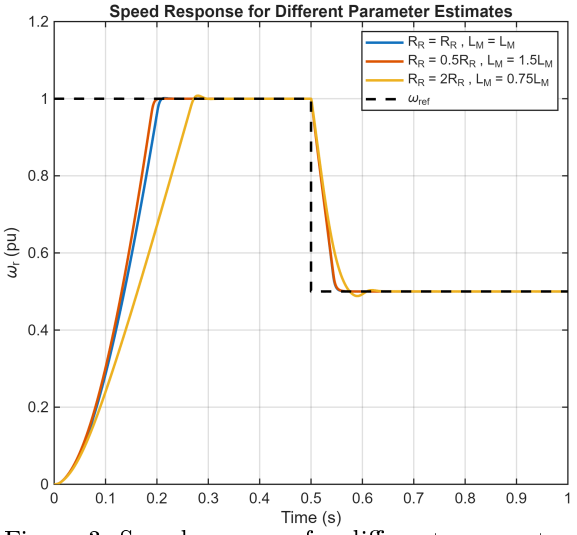


Figure 3: Speed response for different parameter estimates

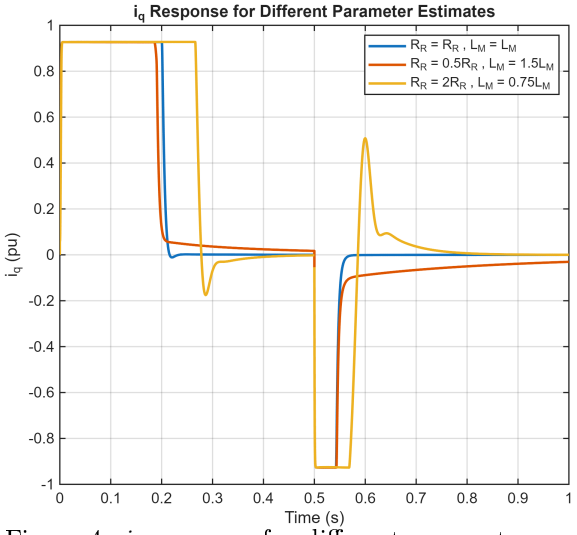


Figure 4: i_q response for different parameter estimates

The results show that using the actual parameter values, namely $\widehat{R}_R = R_R$ and $\widehat{L}_M = L_M$, gives more accurate field orientation because ψ_q remains very close to zero in steady state. This means that the d axis remains correctly aligned with the rotor field.

In contrast, incorrect parameter estimates cause a significant orientation error, making ψ_q deviate from zero and producing larger variations and oscillations in ψ_d.

The speed response also becomes slower when the current-model parameters deviate from the actual machine values. This occurs because the flux-estimation error directly affects slip-frequency calculation and therefore indirect field orientation. Consequently, the decoupling between flux and torque deteriorates, impairing the dynamic behaviour of the speed-control loop.

Finally, incorrect parameters produce larger variations and overshoots in i_q, as the speed controller attempts to compensate for the orientation error and reduced electromagnetic-torque capability.

2 Modified Current Model

In this question, the modified current model was implemented to reduce the sensitivity of indirect field orientation to estimation errors in rotor resistance R_R and magnetizing inductance L_M.

In the conventional current model, the orientation angle is calculated exclusively from the estimated motor parameters. Parameter errors therefore degrade vector-control performance.

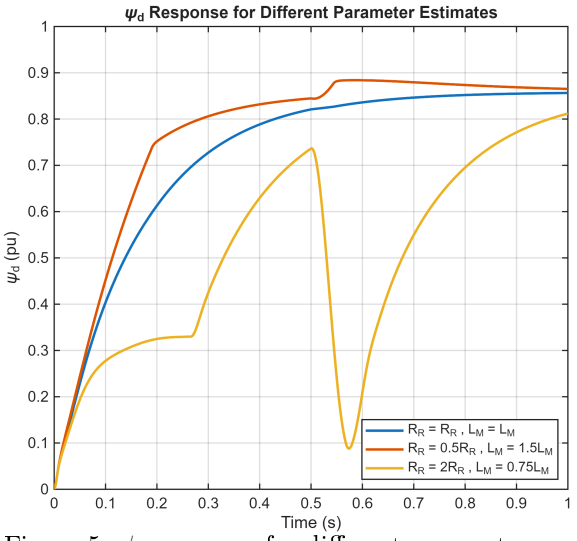


Figure 5: ψ_d response for different parameter estimates

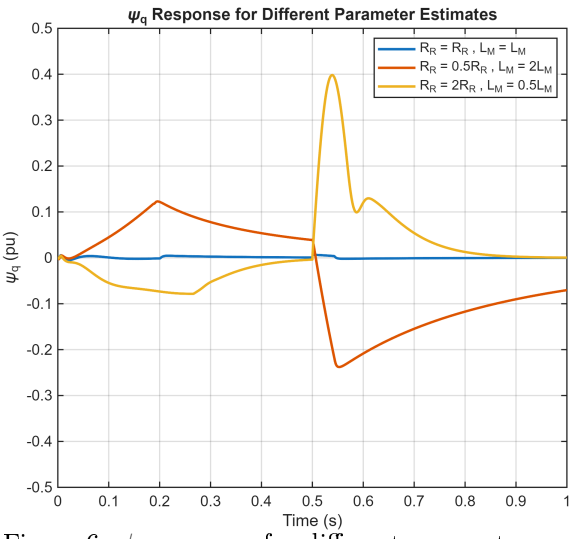


Figure 6: ψ_q response for different parameter estimates

The modified current model introduces additional compensation terms to limit the effects of estimation errors and maintain more accurate field orientation, as described by the following equations:

$$\widehat{E}_d = \bar{v}_d^{ref} - \widehat{R}_s i_d^{ref} + \omega_1 \widehat{L}_\sigma i_q^{ref} \tag{7}$$

$$\lambda_s = \mathrm{sat}\left(\lambda \frac{\omega_r}{\omega_\Delta}, \lambda\right) \tag{8}$$

$$\omega_1 = \frac{a_l}{s + a_l}\left(\omega_r + \frac{\widehat{R}_R i_q^{ref} - \lambda_s \widehat{E}_d}{\widehat{\psi}_R}\right) \tag{9}$$

$$\theta_1 = \mathrm{mod}\left(\int \omega_1 \, dt, 2\pi\right) \tag{10}$$

$$\widehat{\psi}_R = \int \widehat{R}_R \left(i_d^{ref} - \frac{\widehat{\psi}_R}{\widehat{L}_M}\right) dt \tag{11}$$

Compared with the conventional current model, the modified model introduces an additional correction to the synchronous angular-speed estimate through the voltage error $\widehat{E}_d$. This correction compensates for errors caused by inaccurate estimates of $\widehat{R}_R$ and $\widehat{L}_M$, significantly improving flux-estimation accuracy.

The correction is also weighted by λ_s, obtained from a saturated ramp function. Its influence is thus reduced at low speeds, where the system dynamics are more sensitive and full activation could cause instability by changing the roots of the characteristic polynomial. As speed increases, λ_s approaches its maximum and the correction becomes fully active.

Figure (7) shows the overall system, while Figure (8) shows the modified current-model subsystem.

Simulation Results

Three simulations were performed to investigate the modified model, using the same parameter estimates as in the preceding question.

Figures (9)–(12) show the speed response, the current component i_q and the rotor-flux components in the dq reference frame.

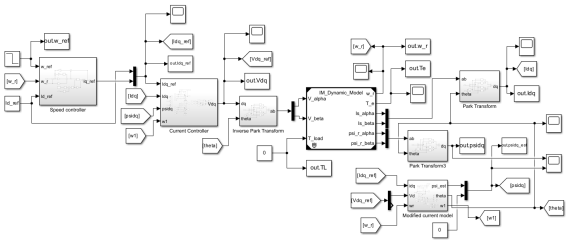


Figure 7: Complete IM model

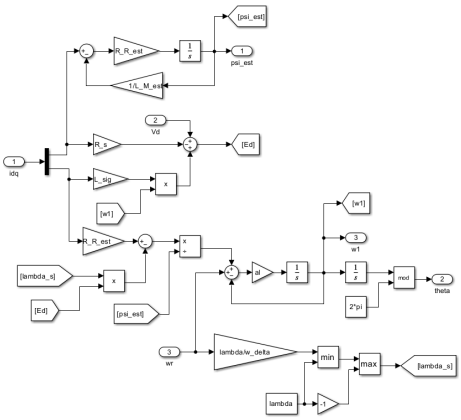


Figure 8: Modified current model subsystem

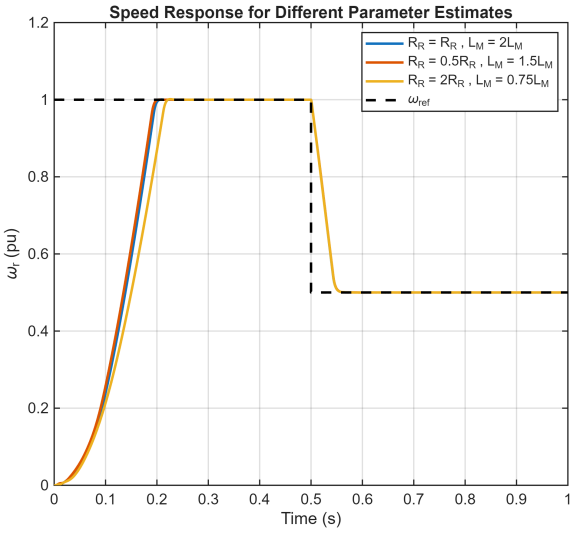


Figure 9: Speed response for different parameter estimates

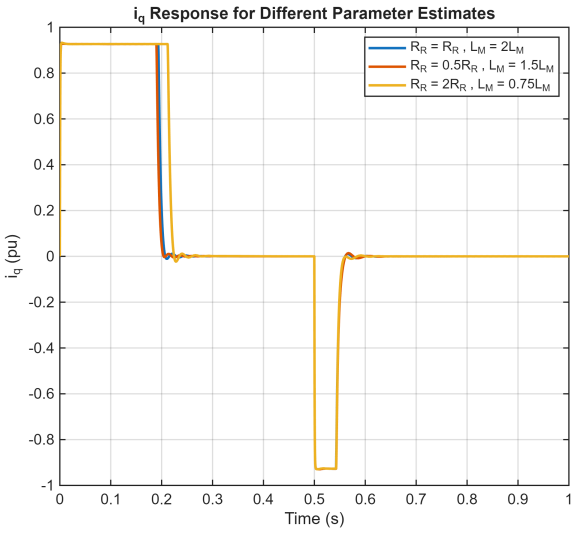


Figure 10: i_q response for different parameter estimates

The results show that the modified current model is much less sensitive to parameter-estimation errors than the conventional model of the preceding question. In particular, simulations with incorrect parameters behave almost identically to the case with perfect estimates.

In this case, even with correctly estimated machine parameters, the q-axis flux ψ_q is not zero during motor-speed transients. This occurs because the modified current model does not directly align the reference frame with the estimated rotor-flux vector; instead, it uses a synchronous angular-speed correction mechanism based on the voltage error $\widehat{E}_d$. Sudden speed changes produce a small estimation error, causing a momentary deviation between the actual and estimated flux angles.

3 Operation in the Field-Weakening Region

In this question, a field-weakening controller was implemented to enable machine operation above rated speed.

In the base operating region, rotor flux is kept constant at its rated value to make full use of the available torque-producing capability. As speed increases, however, the required stator voltage

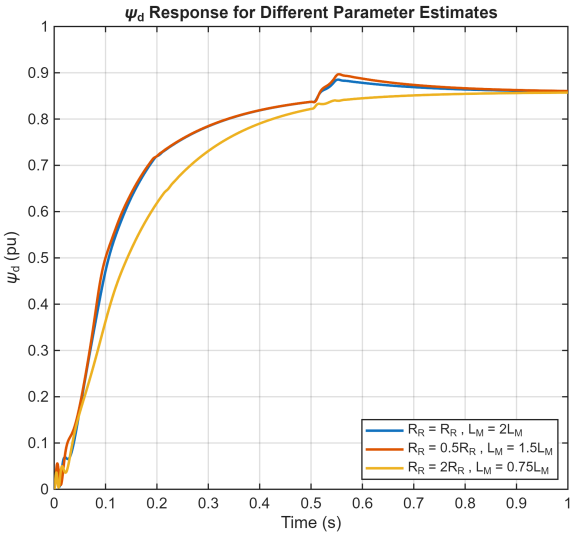


Figure 11: ψ_d response for different parameter estimates

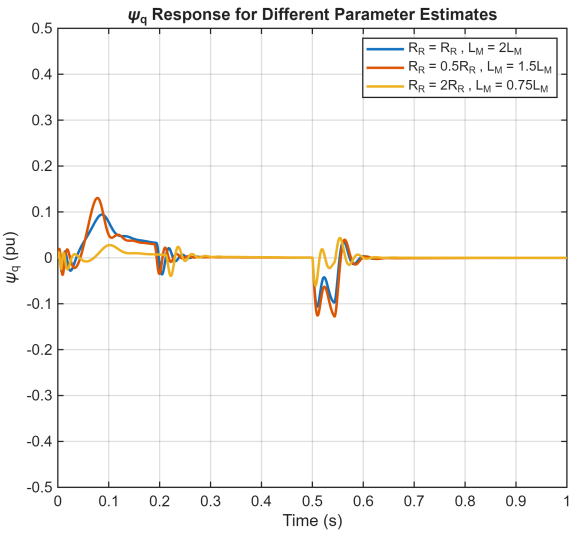


Figure 12: ψ_q response for different parameter estimates

increases with operating frequency. Since the converter has a finite maximum output voltage, rated flux cannot be maintained indefinitely.

Field weakening is therefore applied when the required stator voltage approaches the available inverter limit. In this region, the magnetizing-current component is reduced to limit the required voltage and permit motor operation above rated speed.

The field-weakening controller is described by the following relationships:

$$k_{fw} = \frac{\widehat{R}_R}{\widehat{L}_\sigma^2 V_{base} \max(|\omega_1|, \omega_{base})} \tag{12}$$

$$i_d^{ref} = k_{fw} \int \left[V_{base}^2 - (v_d^{ref})^2 - (v_q^{ref})^2 + \frac{1}{k_{fw}} (i_{d,sat}^{ref} - i_d^{ref}) \right] dt \tag{13}$$

$$i_{d,sat}^{ref} = \min\Big[\max(i_d^{ref}, 0.1 I_{d,nom}), I_{d,nom}\Big] \tag{14}$$

$$\widehat{\xi} = \frac{\widehat{L}_\sigma + \widehat{L}_M}{\widehat{L}_\sigma} \tag{15}$$

$$i_q^{ref} = \text{sat}\left[i_{q,\text{nom}}^{ref}, \min\left(\sqrt{I_{max}^2 - (i_d^{ref})^2}, \widehat{\xi}\, i_d^{ref} \right) \right] \tag{16}$$

When the required voltage approaches the inverter limit, i_d^{ref} is gradually reduced, decreasing rotor flux. This limits the stator-voltage magnitude and enables operation above rated speed without exceeding the system voltage and current constraints.

The product $\omega_r \psi_R$ thus remains approximately constant, keeping the required voltage within the converter limits. Operation above rated speed becomes possible at the cost of reduced available electromagnetic torque.

Figure (13) shows the overall simulation model with the field-weakening controller added. Figure (14) shows the field-weakening subsystem used for above-rated-speed operation.

The implementation uses the modified current model from the preceding question. The speed reference is selected to drive the machine to an operating region reaching five times rated speed.

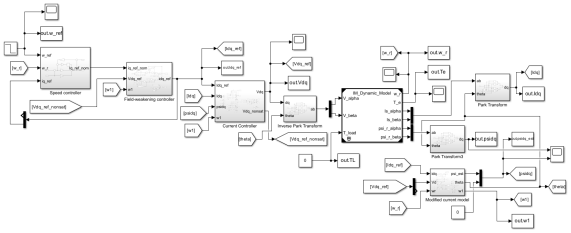


Figure 13: Complete IM model

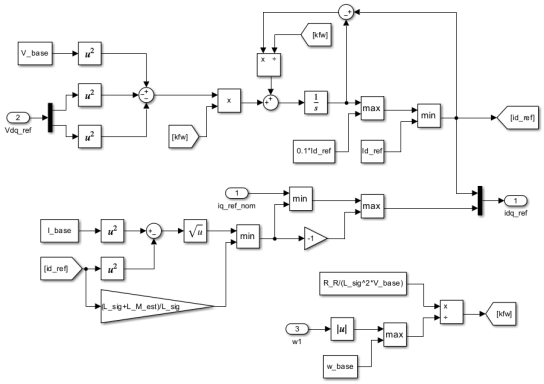


Figure 14: Field-weakening controller subsystem

Simulation Results

Figures (15)–(18) show the speed response, electromagnetic torque, stator currents and rotor-flux components during field-weakening operation.

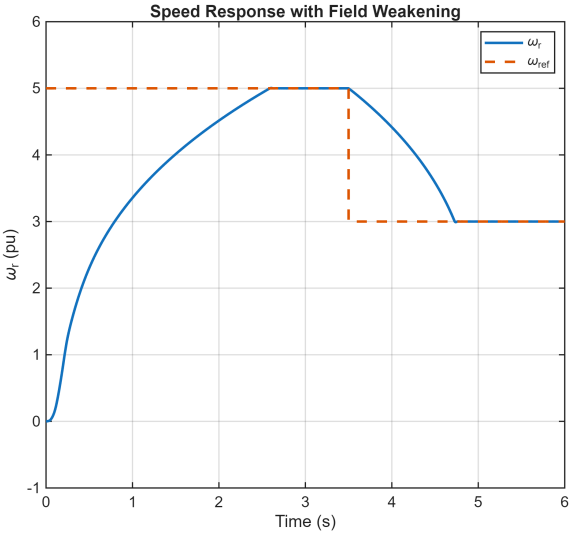


Figure 15: Speed response with field weakening

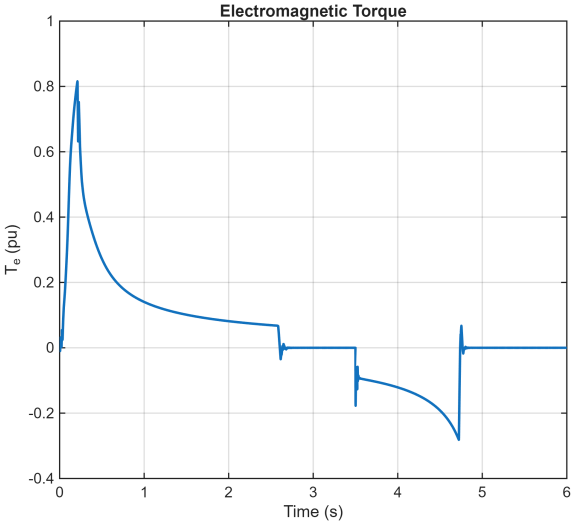


Figure 16: Electromagnetic torque

The results show that rotor speed successfully follows the reference even when it exceeds the rated value. This confirms that the field-weakening controller permits operation in the supersynchronous-speed region without violating the converter voltage constraints.

As speed rises above the rated value, the magnetizing-current component i_d decreases to reduce rotor flux. This reduction is also visible in the flux waveforms, where ψ_d decreases upon entering the field-weakening region.

Electromagnetic torque remains sufficient to accelerate the machine, but its available value is lower than in the base operating region. This is expected because reducing flux correspondingly reduces the maximum torque that can be developed.

Finally, apart from small transient variations, ψ_q remains very close to zero throughout the simulation. This indicates that field orientation is maintained with satisfactory accuracy even in the field-weakening region. Consequently, combining the modified current model with the field-weakening controller enables stable operation above rated speed.

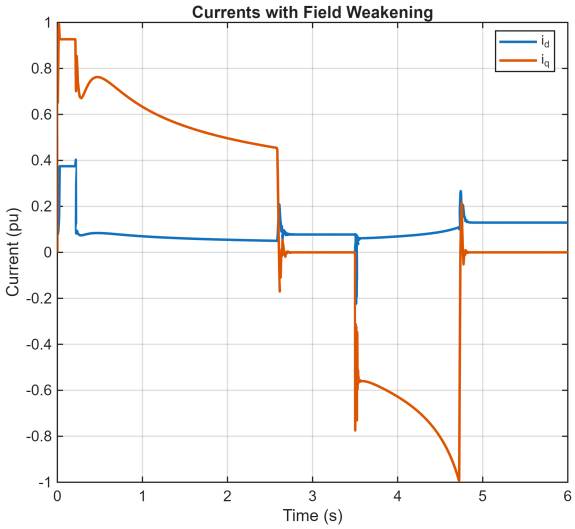


Figure 17: Stator currents

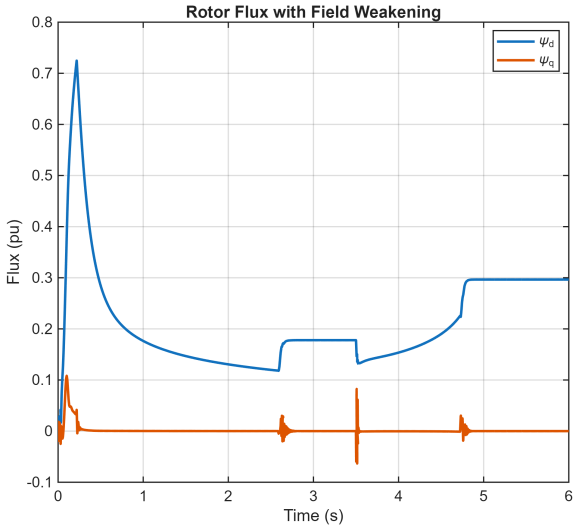


Figure 18: Rotor flux components

4 Comparison of Conventional and Modified Current Models in the Field-Weakening Region

The final question compares the conventional and modified current models during field-weakening operation in the presence of a rotor-resistance estimation error. Specifically, the following case is examined:

$$\widehat{R}_R = 2R_R \tag{17}$$

Magnetizing inductance is assumed to be known accurately:

$$\widehat{L}_M = L_M \tag{18}$$

The comparison aims to assess how rotor-resistance error affects flux estimation and field orientation above rated speed. In the field-weakening region, rotor flux decreases and slip frequency increases, making the system more sensitive to parameter-estimation errors.

Four simulations were therefore performed: the conventional and modified current models, each with correct and incorrect rotor-resistance estimates.

Simulation Results

Figures (19)–(24) show the speed response, electromagnetic torque, rotor-flux components and stator currents for the different simulation cases.

With correct parameter estimates, the modified and conventional current models exhibit almost identical speed and torque responses. The only substantial difference is in ψ_q, where the modified model shows a larger momentary deviation from zero during transients, for the reasons discussed in Question 2.

The modified model with an incorrect rotor-resistance estimate follows the preceding cases very closely, with practically imperceptible slowing of the response. System dynamics remain smooth, without a significant performance difference during the transition into the field-weakening region.

In contrast, the conventional current model with an incorrect rotor-resistance estimate exhibits considerably worse dynamic behaviour. The speed response is slower, and strong oscillations appear in the flux components during the transition from $5\omega_{base}$ to $3\omega_{base}$. These oscillations are caused by the field-orientation error, which is amplified in the field-weakening region by the speed-reference change.

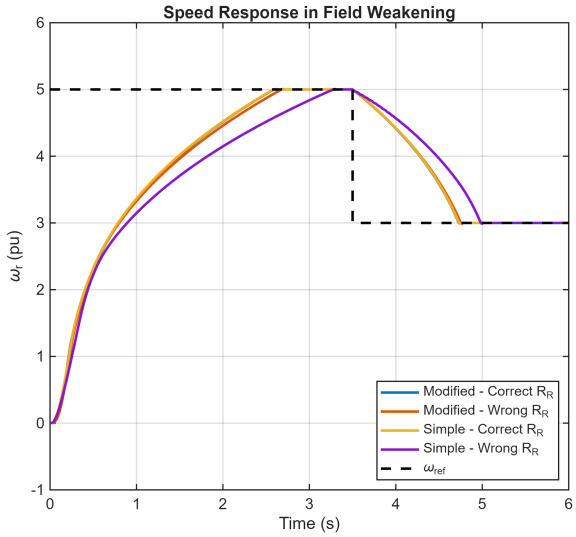


Figure 19: Speed response in field weakening

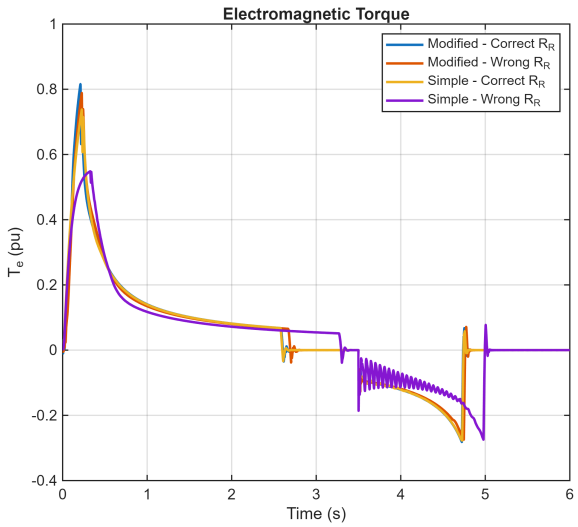


Figure 20: Electromagnetic torque

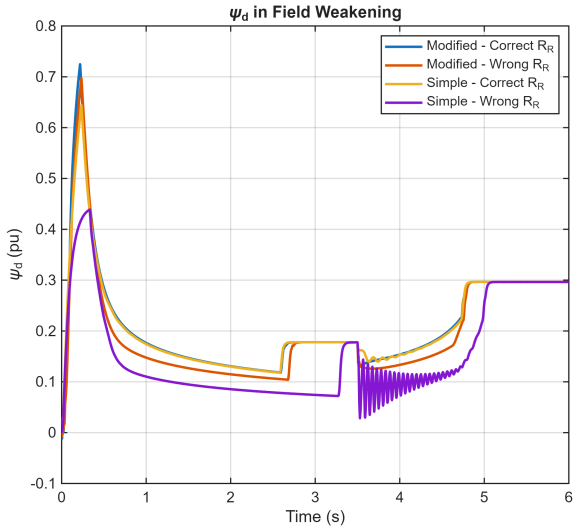


Figure 21: ψ_d component

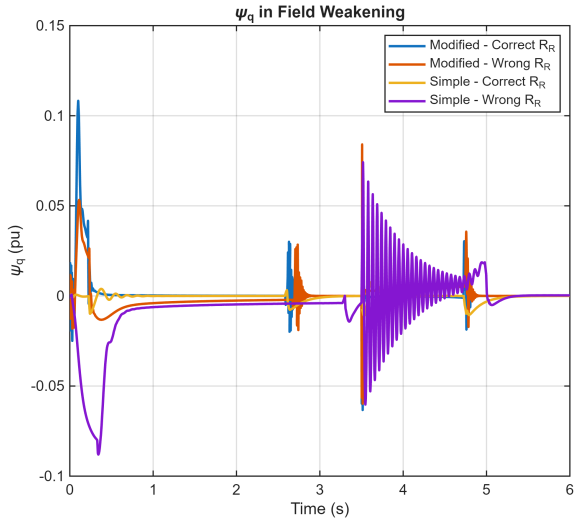


Figure 22: ψ_q component

Consequently, flux disturbances are transferred directly to the stator currents, causing corresponding large oscillations in i_d and i_q, ultimately producing a degraded, oscillatory electromagnetic-torque response.

The larger error in the conventional model arises because synchronous angular-speed and slip estimation depend directly on $\widehat{R}_R$. When this parameter is incorrect, estimated slip deviates significantly from actual slip, leading to incorrect orientation of the dq reference frame relative to rotor flux. In the field-weakening region, where flux is already reduced and torque is produced with a smaller stability margin, the system is more sensitive to such errors. Small deviations in flux estimation therefore translate into large current deviations and, ultimately, electromagnetic-torque oscillations.

Overall, the conventional current model is shown to be highly sensitive to rotor-resistance errors in the field-weakening region, as the orientation error is dynamically amplified and affects all control loops. In contrast, the modified model significantly reduces this sensitivity through its corrected synchronous angular-speed estimate, maintaining stable, reliable operation even under substantial

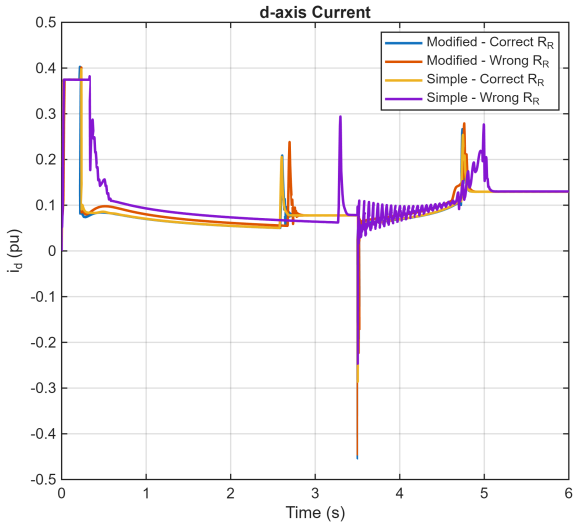


Figure 23: i_d current

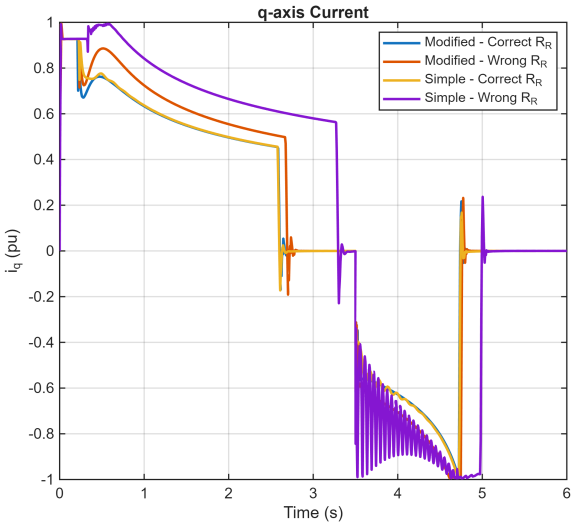


Figure 24: i_q current

parameter deviations.